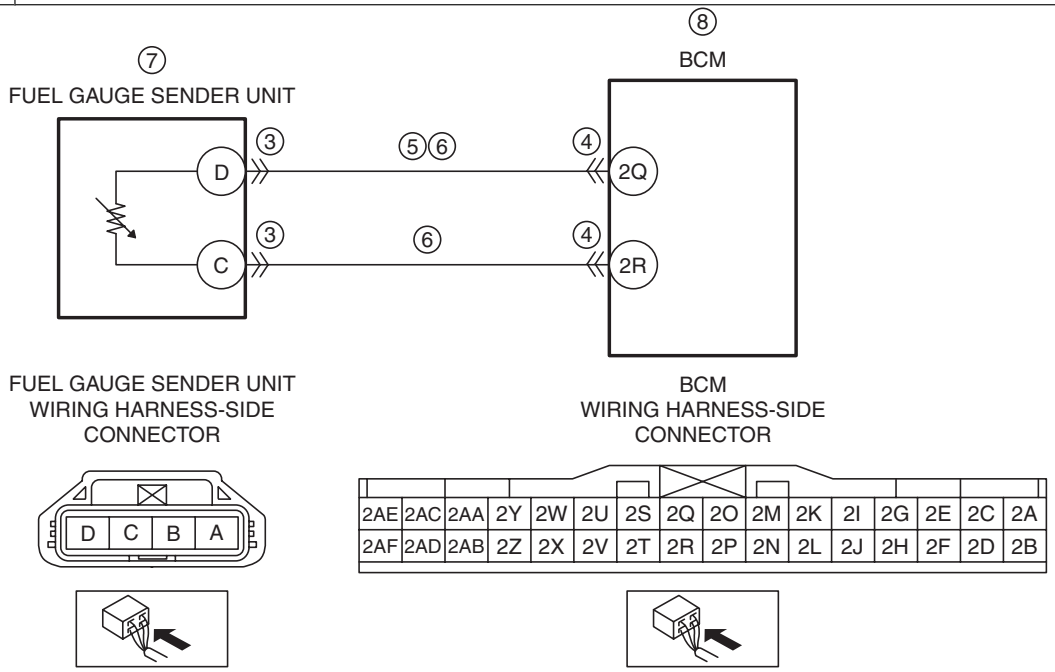
DTC P0463:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0463:00	Fuel gauge sender unit circuit high input
DETECTION CONDITION	• The PCM monitors the fuel level signal and fuel gauge sender unit output voltage from the instrument cluster. If the PCM detects a fuel level or fuel gauge sender unit output voltage that is too high, the PCM determines that the fuel gauge sender unit circuit has a malfunction. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Fuel gauge sender unit connector or terminals malfunction • BCM connector or terminals malfunction • Short to power supply in wiring harness between fuel gauge sender unit terminal D and BCM terminal 2Q • Open circuit in wiring harness between the following terminals: — Fuel gauge sender unit terminal D—BCM terminal 2Q — Fuel gauge sender unit terminal C—BCM terminal 2R • Fuel gauge sender unit malfunction • BCM malfunction • Instrument cluster malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT FUEL GAUGE SENDER UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fuel gauge sender unit connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
4	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
5	**INSPECT FUEL GAUGE SENDER UNIT SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the fuel gauge sender unit and BCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the fuel gauge sender unit terminal D (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuel gauge sender unit terminal D and BCM terminal 2Q. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
6	**INSPECT FUEL GAUGE SENDER UNIT CIRCUIT FOR OPEN CIRCUIT** • Verify that the fuel gauge sender unit and BCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Fuel gauge sender unit terminal D—BCM terminal 2Q — Fuel gauge sender unit terminal C—BCM terminal 2R • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Fuel gauge sender unit terminal D—BCM terminal 2Q • Fuel gauge sender unit terminal C—BCM terminal 2R **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
7	**INSPECT FUEL GAUGE SENDER UNIT** • Inspect the fuel gauge sender unit. (See FUEL GAUGE SENDER UNIT INSPECTION.) • Is there any malfunction?	Yes	Replace the fuel gauge sender unit, then go to Step 10. (See FUEL PUMP UNIT DISASSEMBLY/ASSEMBLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
8	**INSPECT BCM** • Inspect the BCM. (See BODY CONTROL MODULE (BCM) INSPECTION.) • Is there any malfunction?	Yes	Replace the BCM, then go to Step 10. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Go to the next step.
9	**INSPECT INSTRUMENT CLUSTER** • Inspect the instrument cluster. (See INSTRUMENT CLUSTER INSPECTION.) • Is there any malfunction?	Yes	Replace the instrument cluster, then go to the next step. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
		No	Go to the next step.

STEP	INSPECTION		ACTION
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.